

Heme B Biosynthesis in *Pseudomonas putida* KT2440 — Module/Pathway/Taxon Review

Review topic: Heme B biosynthesis through alternative entry and oxidation routes in *Pseudomonas putida* KT2440 **Target taxon:** PSEPK — *Pseudomonas putida* KT2440 (NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu00860` "Porphyrin metabolism" (resolved local bucket, 45–46 primary candidate genes; module area *cofactors_vitamins_redox*)

1. Executive Summary

Heme B biosynthesis is **fully satisfiable** in *Pseudomonas putida* KT2440. Every catalytic step from 5-aminolevulinate (ALA) formation through ferrochelation of protoporphyrin IX to heme B is encoded in the proteome by a clearly identifiable candidate gene. The organism realizes one specific combination of the alternative routes that the generic module allows: it forms ALA exclusively via the **C5 / glutamyl-tRNA route** (`hemA`, `hemL`, with `gltX` supplying charged Glu-tRNA); it carries **both** the oxygen-dependent (`hemF`) and oxygen-independent (`hemN`) coproporphyrinogen III oxidases, giving aerobic/anaerobic flexibility; it uses a single **HemJ-type protoporphyrinogen IX oxidase** (`PP_0431`); and it converts protoporphyrin IX to heme B with a single-copy **ferrochelatase** (`hemH`). The Shemin/C4 ALA synthase and the HemG and HemY protoporphyrinogen-oxidase families are **not expected** in this organism, confirmed by a proteome-wide Pfam scan (HemJ PF03653 = 1 hit; true HemG PF01903 = 0; no HemY-fold protein is a protoporphyrinogen oxidase).

The single most important curation finding is one of **scope**, not gaps. The KEGG `ppu00860` "Porphyrin metabolism" bucket is an umbrella map: only ~13 of the 46 candidate genes belong to the heme B trunk. The remaining ~33 candidates route to neighboring tetrapyrrole branches that should be kept in separate modules — cobalamin (B₁₂) biosynthesis (the large `cob*` / `cbiD` / `pdu0` block), siroheme synthesis (`cysG`, `cobA` / `PP_2090`, `PP_0188`), heme catabolism and iron storage/acquisition (`hemO`, `PP_2582`, `PP_1358`, bacterioferritins, ferroxidase), and downstream heme modification for terminal oxidase cofactors (`cyoE1` / `cyoE2` heme O synthases and `PP_0109`, which is a mislabeled CtaA/heme A synthase, **not** a "Cytochrome B").

The two curation actions that require follow-up at the gene level are: (i) two porphobilinogen-synthase (ALAD) paralogs, `hemB` (PP_2913) and `hemBB` (PP_3322), which are annotated identically and need metal-cofactor/subtype disambiguation; and (ii) the generically annotated `PP_0431`, which should be curated explicitly to the HemJ route. All conclusions rest on UniProt family/domain evidence for the target strain plus well-established biochemical literature for the pathway; direct genetic/enzymatic experiments in KT2440 itself remain the principal evidence gap.

2. Target-Organism Pathway Definition

2.1 What process is included

The module covers **protoporphyrin-dependent heme B biosynthesis**: the linear conversion of the universal tetrapyrrole precursor 5-aminolevulinate through the shared "trunk" (porphobilinogen → hydroxymethylbilane → uroporphyrinogen III → coproporphyrinogen III), then the two late oxidation steps (coproporphyrinogen III → protoporphyrinogen IX → protoporphyrin IX), and finally ferrous-iron insertion (ferrochelation) to yield heme B (protoheme IX). Upstream, the module includes the *route-specific* formation of ALA, for which two mutually exclusive entries exist across taxa: the C5 (glutamyl-tRNA) route and the C4 (Shemin, ALA synthase) route.

2.2 Neighboring pathways to keep separate

For curation, the following processes that co-occur in the KEGG `ppu00860` umbrella map should be **excluded** from the heme B module:

- **Cobalamin (vitamin B₁₂) biosynthesis** — the aerobic corrin ring pathway (`cobA`/SUMT, `cobI`, `cobG`, `cobJ`, `cobM`, `cobF`, `cobK`, `cobL`, `cbiD`, `cobH`, `cobB`, `cobN`, `cobO`, `cobQ`, `cobC`, `cobD`, `cobP`, `cobT`, `cobS`, `PP_1680`, `PP_3409`, `PP_3506`, `pdu0`). These branch from uroporphyrinogen III via precorrin-2, sharing only the very earliest trunk with heme.
- **Siroheme biosynthesis** — `cysG` (multifunctional siroheme synthase), and the SUMT activities of `cobA`/PP_2090 and `PP_0188`. Siroheme also branches at uroporphyrinogen III.
- **Heme catabolism / iron acquisition and storage** — heme oxygenases (`hem0`/PP_1005, `PP_2582`), heme iron utilization protein (`PP_1358`), bacterioferritins (`bfr-I`/PP_0482, `bfr-`

II /PP_1082), and ferroxidase (PP_4856). These consume heme or store iron and are downstream/orthogonal to biosynthesis.

- **Heme O / heme A modification for terminal oxidases** — protoheme IX farnesyltransferases / heme O synthases cyoE1 (PP_0110) and cyoE2 (PP_0816), and the heme A synthase PP_0109 (CtaA/COX15 family). These act *on* finished heme B to make oxidase cofactors and belong to a "heme modification for respiration" module, not heme B biosynthesis.

2.3 Alternate names / database definitions

The pathway is variously called "heme biosynthesis," "protoheme IX biosynthesis," "tetrapyrrole biosynthesis (heme branch)," and MetaCyc "heme *b* biosynthesis I (aerobic)/II (anaerobic)". KEGG collapses all tetrapyrrole end-products (heme, siroheme, cobalamin, and their catabolism) into the single map 00860 "Porphyrin metabolism" (historically "Porphyrin and chlorophyll metabolism"), which is why the local bucket is far broader than heme B proper.

3. Expected Step Model and Satisfiability

The table below maps each generic module step to the KT2440 gene(s) and a satisfiability call.

Module step	Enzyme (EC)	KT2440 gene (locus UniProt)	Call
ALA entry — C5	Glutamyl-tRNA synthetase (6.1.1.17)	gltX (PP_1977 Q88LF6)	covered
	Glutamyl-tRNA reductase (1.2.1.70)	hemA (PP_0732 Q88PW6)	covered
	Glutamate-1-semialdehyde 2,1-aminomutase (5.4.3.8)	hemL (PP_4784 Q88DP0)	covered
ALA entry — C4/ Shemin	5-aminolevulinate synthase (2.3.1.37)	<i>absent</i>	not_expected_in_target_taxon
ALA condensation	Porphobilinogen synthase / ALAD (4.2.1.24)	hemB (PP_2913 Q88IT6) and hemBB (PP_3322 Q88HN1)	covered (paralog ambiguity)
Hydroxymethylbilane formation	Porphobilinogen deaminase / HMBS (2.5.1.61)	hemC (PP_0186 Q88RE5)	covered
Uroporphyrinogen III formation	Uroporphyrinogen-III synthase (4.2.1.75)	hemD (PP_0187 Q88RE4)	covered
Coproporphyrinogen III formation	Uroporphyrinogen decarboxylase (4.1.1.37)	hemE (PP_5074 Q88CV6)	covered
Coprogen oxidation — O ₂ -dependent	Coproporphyrinogen oxidase HemF (1.3.3.3)	hemF (PP_0073 Q88RQ6)	covered
			covered

Module step	Enzyme (EC)	KT2440 gene (locus UniProt)	Call
Coprogen oxidation — O ₂ -independent	Coproporphyrinogen dehydrogenase HemN (1.3.98.3)	hemN (PP_4264 Q88F35)	
Protoporphyrinogen IX oxidation — HemJ	HemJ-family Protox	PP_0431 (Q88QQ7)	covered (HemJ route)
Protoporphyrinogen IX oxidation — HemG	Quinone-acceptor Protox	absent	not_expected_in_target_taxon
Protoporphyrinogen IX oxidation — HemY	O ₂ -acceptor PPOX	absent	not_expected_in_target_taxon
Ferrochelation	Ferrochelatase HemH (4.98.1.1)	hemH (PP_0744 Q88PV4)	covered

Every trunk step is covered. The only step-level uncertainties are the *choice of variant* at the ALA-entry, coprogen-oxidation, and protoporphyrinogen-oxidation nodes — which this review resolves — plus the ALAD paralog ambiguity.

4. Candidate Genes and Evidence

4.1 ALA entry: the C5 (glutamyl-tRNA) route (Finding F002)

P. putida KT2440 forms ALA via the **C5 route**, not the C4/Shemin route. The candidate list encodes glutamyl-tRNA reductase **hemA** (PP_0732, Q88PW6, EC 1.2.1.70), glutamate-1-semialdehyde 2,1-aminomutase **hemL** (PP_4784, Q88DP0, EC 5.4.3.8), and glutamate-tRNA ligase **gltx** (PP_1977, Q88LF6, EC 6.1.1.17). No 5-aminolevulinate synthase (ALAS, EC 2.3.1.37) gene is present anywhere in the bucket. This is the canonical bacterial arrangement: in the C5 pathway Glu-tRNA "is transformed to delta-aminolevulinic acid, the universal precursor of tetrapyrroles (e.g., heme and chlorophyll) by the action of Glu-tRNA reductase (GluTR) and glutamate semialdehyde aminotransferase" (PMID: 17360620). Arabidopsis and *E. coli* work

confirms ALA is derived from glutamate via the two-step C5 pathway (HemA/GluTR + GSA1/GSAM) (PMID: 7908550). The C4/ALAS route is largely restricted to α -Proteobacteria, animals, and fungi, so its absence in this γ -proteobacterium is expected, not a gap.

Curation caveat: `gltX` is a **shared/dual-function** gene — the charged Glu-tRNA it produces feeds both protein synthesis and tetrapyrrole biosynthesis (PMID: 17360620). It should be flagged as contributing to, but not exclusive to, the heme module.

4.2 Shared tetrapyrrole trunk (Findings F003, F004)

All shared-trunk enzymes are present as single, family-verified copies (except ALAD, below):

- `hemC` (Q88RE5) — porphobilinogen deaminase / HMBS family, dipyrromethane cofactor.
- `hemD` (Q88RE4) — uroporphyrinogen-III synthase family.
- `hemE` (Q88CV6) — uroporphyrinogen decarboxylase (URO-D) family.

The *B. subtilis* `hemAXCDBL` operon studies confirm these genes constitute the conserved glutamate-to-uroporphyrinogen-III core and that `hemA`, `hemB`, `hemC`, `hemD` have all been shown to be essential for heme synthesis (PMID: 1672867; PMID: 2110138), supporting the expectation that their KT2440 orthologs are functional and required.

4.3 Two ALAD (porphobilinogen synthase) paralogs (Finding F005)

KT2440 carries **two** ALAD paralogs: `hemB` (PP_2913, Q88IT6, 324 aa) and `hemBB` (PP_3322, Q88HN1, 333 aa). Both belong to the ALAD family (Pfam PF00490, EC 4.2.1.24) and are annotated identically as delta-aminolevulinic acid dehydratase; UniProt lists no explicit metal cofactor for either. Bacterial ALADs split into **Zn²⁺-dependent** (cysteine-rich active-site motif) and **Mg²⁺-dependent** (aspartate motif) subtypes with distinct physiology. Most *Pseudomonas* genomes encode a single `hemB`, so this duplication is notable and biologically informative — the two paralogs may differ in metal use, expression condition, or physiological role. This is the clearest gene-level curation ambiguity in the module.

4.4 Coproporphyrinogen III oxidation: both HemF and HemN (Findings F003, F004)

KT2440 encodes **both** coproporphyrinogen III oxidases: the oxygen-dependent `hemF` (PP_0073, Q88RQ6, aerobic CPO, EC 1.3.3.3) and the oxygen-independent radical-SAM `hemN` (PP_4264, Q88F35, EC 1.3.98.3). Possessing both isozymes is the expected arrangement for a metabolically

versatile soil pseudomonad, enabling heme synthesis across aerobic and microaerobic/anaerobic conditions. Both nodes of the "coproporphyrinogen oxidation chemistry" alternative are therefore realized.

4.5 Protoporphyrinogen IX oxidation: HemJ only (Findings F001, F004)

The penultimate step is catalyzed by a **HemJ-type** protoporphyrinogen IX oxidase, **PP_0431** (Q88QQ7). UniProt annotates it generically as "Protoporphyrinogen IX oxidase (EC 1.3.99.-)", but it carries Pfam **PF03653** and InterPro "HemJ-like" and is placed by UniProt in the HemJ family. A proteome-wide scan of UP000000556 found **PF03653 (HemJ) = 1 hit** (PP_0431), **PF01903 (true HemG) = 0 hits**, and while PF01593 (HemY/amine-oxidase fold) returned 5 hits, all are renalase/tryptophan-2-monooxygenase/amine oxidases — **none a protoporphyrinogen oxidase**. KT2440 therefore has no HemG or HemY redundancy at this step.

This is mechanistically notable. "Three nonhomologous isofunctional enzymes, HemG, HemJ, and HemY, for Protox have been identified" and most organisms possess only one type; "HemG is mostly limited to γ -Proteobacteria whereas HemJ may have originated within α -Proteobacteria and transferred to other Proteobacteria" ([PMID: 25108393](#)). *Pseudomonas* is a γ -proteobacterium — the "expected" home of HemG — yet KT2440 uses HemJ, a clear example of the intricate, mosaic evolutionary distribution of these enzymes. Heterologous complementation work confirms that HemJ, HemY, and HemG homologs are genuinely interchangeable in function despite being unrelated in sequence ([PMID: 36357749](#)), which validates assigning PP_0431 to the Protox step on family evidence.

4.6 Ferrochelation (Finding F004)

hemH (PP_0744, Q88PV4) is a **single-copy** ferrochelatase (family-verified, EC 4.98.1.1) that inserts Fe^{2+} into protoporphyrin IX to form heme B. This is the species-neutral convergence point of the module and is unambiguously covered.

4.7 Genes that belong to neighboring modules (over-scoped in the bucket)

Candidate(s)	True branch	Recommended module
cobA /PP_2090, cobI, cobG, cobJ, cobM, PP_3763 (CobF), cobK, cobL, cbiD, cobH, cobB, cobN, PP_3506, cobO, cobQ, cobC, cobD, cobP, cobT, cobS, PP_1680, PP_3409 (CobE), pdu0	Cobalamin (B ₁₂) biosynthesis	separate cobalamin module
cysG, PP_0188, SUMT activity of cobA	Siroheme biosynthesis	separate siroheme module
hemO (PP_1005), PP_2582, PP_1358	Heme catabolism / iron utilization	heme-degradation module
bfr-I, bfr-II, PP_4856	Iron storage / ferroxidase	iron-homeostasis module
cyoE1 (PP_0110), cyoE2 (PP_0816)	Heme O synthase (EC 2.5.1.141)	heme-O/A modification module
PP_0109	Heme A synthase (CtaA/COX15 family) — mislabeled "Cytochrome B"	heme-O/A modification module

5. Gaps, Ambiguities, and Likely Over-Annotations

1. **PP_0109 is mis-annotated.** It is labeled "Cytochrome B" in the metadata, but its InterPro/ family signature (HemeA_synthase_prok; CtaA/COX15) identifies it as a **heme A synthase** that converts heme O to heme A. It is not a cytochrome b apoprotein and does not belong to the heme B trunk. **Correct this annotation and move it to the heme-modification module.**

2. **PP_0431** carries an over-broad EC and generic name. "Protoporphyrinogen IX oxidase (EC 1.3.99.-)" understates the specific, curatable assignment: it is a **HemJ-family** Protox. Promote to explicit HemJ annotation.
3. **ALAD paralog ambiguity (hemB vs hemBB)**. Two identically annotated PF00490 proteins with no listed metal cofactor. Their Zn^{2+} - vs Mg^{2+} -dependence, conditional expression, and which (if either) is the housekeeping enzyme are unresolved.
4. **Broad/duplicated EC and GO mappings inflating the bucket**. Several **cob*** genes and **cysG** carry SUMT/uroporphyrinogen-III C-methyltransferase (EC 2.1.1.107) activity that overlaps the earliest trunk, which can cause over-propagation of "porphyrin metabolism" into heme B satisfiability checks. The bacterioferritins and **PP_4856** carry EC 1.16.3.1 (ferroxidase) unrelated to biosynthesis.
5. **gltX shared function**. Contributes charged Glu-tRNA to both translation and tetrapyrrole synthesis; should not be treated as a heme-exclusive gene.
6. **No genuine heme B trunk gaps were found**. The module is complete; there is no missing-step gap requiring a paralog search or a "not found" call for any trunk reaction.

6. Module and GO-Curation Recommendations

Step-level calls for the heme B module in PSEPK:

Module step	Call
C5 ALA entry (<code>hemA</code> , <code>hemL</code> , <code>gltX</code>)	covered
C4/Shemin ALA entry (ALAS)	not_expected_in_target_taxon
ALA condensation (ALAD)	covered — flag paralog ambiguity (<code>hemB</code> / <code>hemBB</code>) → candidate_uncertain at gene level
HMBS (<code>hemC</code>), UROS (<code>hemD</code>), URO-D (<code>hemE</code>)	covered
Coprogen oxidation, HemF	covered
Coprogen oxidation, HemN	covered
Protox, HemJ (<code>PP_0431</code>)	covered
Protox, HemG	not_expected_in_target_taxon
Protox, HemY	not_expected_in_target_taxon
Ferrochelation (<code>hemH</code>)	covered

Module boundary: The generic module boundaries are **correct for heme B**, but the *local KEGG bucket* `ppu00860` is too broad — it is an umbrella "Porphyrin metabolism" map. Recommend **splitting** the bucket so that only the ~13 trunk genes populate the heme B module, and creating (or linking to existing) separate modules for cobalamin, siroheme, heme catabolism, iron storage, and heme-O/A modification. Mark the bucket-level relationship as **module_needs_revision** with respect to scope.

GO-term / annotation requests: - Re-annotate `PP_0109` from "Cytochrome B" to **heme A synthase** (CtaA/COX15). - Add explicit **HemJ / protoporphyrinogen oxidase (GO:0004729)** assignment to `PP_0431`. - No new GO term requests appear necessary; existing terms cover all realized steps.

7. Genes to Promote to Full Review (`fetch-gene`)

1. `hemB` (PP_2913, Q88IT6) and `hemBB` (PP_3322, Q88HN1) — disambiguate the two ALAD paralogs (metal-cofactor subtype, physiological role, which is housekeeping). Highest priority.
 2. `PP_0431` (Q88QQ7) — confirm HemJ assignment and replace the generic EC 1.3.99.- annotation.
 3. `PP_0109` (Q88RM0) — correct the "Cytochrome B" mislabel to heme A synthase and remove from the heme B trunk.
 4. `gltX` (PP_1977) — annotate shared/dual role (translation + tetrapyrrole entry).
 5. (Optional) `hemN` (PP_4264) vs `hemF` (PP_0073) — confirm both are functional and note the aerobic/anaerobic redundancy for module logic.
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8. Evidence Base and Limitations

8.1 Key references

PMID	Relevance
25108393	Establishes HemG/HemJ/HemY as three nonhomologous isofunctional Protox enzymes; explains mosaic distribution (HemG mostly γ -Proteobacteria; HemJ from α -Proteobacteria) — underpins the HemJ assignment of PP_0431 (F001).
36357749	Heterologous complementation verifies functional interchangeability of HemJ/HemY/HemG — validates assigning Protox by family.
17360620	Defines C5-route enzymes (GluRS/ gltX , GluTR/ hemA , GSAM/ hemL) and the dual role of Glu-tRNA — underpins F002.
7908550	ALA as universal tetrapyrrole precursor via the two-step C5 pathway (HemA + GSA1) — supports C5 assignment.
1672867	<i>B. subtilis</i> hemAXCDBL cluster; hemA/B/C/D essential for heme — supports trunk-gene functionality by homology.
2110138	Characterization of hemA region; C5 GluTR conserved — supports trunk model.
29164655	<i>Amycolatopsis</i> case with two divergent ALA pathways — cautionary context that some genomes carry dual ALA routes (not the case here).

8.2 Limitations and species-transfer caveats

- **Evidence type is primarily homology/family-based.** All KT2440 assignments rest on UniProt Pfam/InterPro family membership and EC annotation for strain KT2440 proteins, combined with biochemical literature from *other* organisms (*E. coli*, *B. subtilis*, Arabidopsis, cyanobacteria). There are **no direct KT2440 gene-knockout or enzyme-assay data** cited here confirming in-vivo function. Transfer of the biochemical mechanism to KT2440 is **strong** for the highly conserved trunk enzymes and **strong** for the HemJ family assignment (the Pfam signature is diagnostic), but the *physiological* details (e.g., ALAD paralog roles, HemF/HemN condition-specificity) are **uncertain** without strain-specific experiments.
- The proteome scan is a snapshot of UniProt annotations; a novel or divergent Protox not captured by PF03653/PF01903/PF01593 would be missed, though this is unlikely given the complete trunk.

- The ALAD paralog metal-subtype inference is based on general bacterial ALAD biology, not on KT2440 sequence-motif analysis performed here — it should be verified at the sequence level in the `fetch-gene` step.

8.3 Proposed follow-up experiments / actions

1. **Sequence-motif analysis of `hemB` vs `hemBB`** to classify Zn^{2+} - vs Mg^{2+} -dependent ALAD subtypes (active-site Cys-rich vs Asp motif); ideally single/double knockouts to test which supports growth.
 2. **Confirm PP_0431 HemJ function** by heterologous complementation of an *E. coli* ΔhemG strain (as in [PMID: 36357749](#)).
 3. **Correct annotations** for `PP_0109` (heme A synthase) and `PP_0431` (HemJ) in the reference proteome and downstream modules.
 4. **Split the `ppu00860` bucket** into heme B, cobalamin, siroheme, heme-catabolism, iron-storage, and heme-modification modules for satisfiability tracking.
 5. **Test HemF/HemN condition-dependence** (aerobic vs anaerobic growth of single mutants) to confirm the redundancy inferred here.
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9. Mechanistic Model (Realized Route in KT2440)

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Glu --gltX--> Glu-tRNA
      |
      hemA (GluTR)
      v
glutamate-1-semialdehyde
      |
      hemL (GSAM)
      v
[C5 route] 5-aminolevulinate (ALA)      [C4/ALAS: ABSENT]
      |
      hemB / hemBB (ALAD) <-- two paralogs, disambiguate
      v
      porphobilinogen
      |
      hemC (HMBS)
      v
      hydroxymethylbilane
      |
      hemD (UROS)
      v
      uroporphyrinogen III --> [siroheme: cysG; cobalamin: cobA...]
      |
      hemE (URO-D)
      v
      coproporphyrinogen III
      |
      hemF (O2-dep) + hemN (O2-indep) <-- both present
      v
      protoporphyrinogen IX
      |
      PP_0431 (HemJ)      [HemG: ABSENT; HemY: ABSENT]
      v
      protoporphyrin IX
      |
      hemH (ferrochelatase, single copy)
      v
      HEME B
      |
      [cyoE1/cyoE2 -> heme 0; PP_0109 -> heme A] (separate module)

```

Bottom line: The heme B module is satisfiable and complete in *P. putida* KT2440 via C5 entry → shared trunk → HemF/HemN → HemJ → HemH. The work required is curatorial (bucket scoping, three annotation fixes, one paralog disambiguation), not a search for missing biosynthetic capability.

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